

1. AMC 8, 2016, Problem 20

The least common multiple of a and b is 12, and the least common multiple of b and c is 15. What is the least possible value of the least common multiple of a and c ?

2. AMC 8, 2016, Problem 24

The digits 1, 2, 3, 4, and 5 are each used once to write a five-digit number $PQRST$. The three-digit number PQR is divisible by 4, the three-digit number QRS is divisible by 5, and the three-digit number RST is divisible by 3. What is P ?

3. AMC 8, 2018, Problem 21

How many positive three-digit integers have a remainder of 2 when divided by 6, a remainder of 5 when divided by 9, and a remainder of 7 when divided by 11?

4. AMC 8, 2017, Problem 24

Mrs. Sanders has three grandchildren, who call her regularly. One calls her every three days, one calls her every four days, and one calls her every five days. All three called her on December 31, 2016. On how many days during the next year did she not receive a phone call from any of her grandchildren?

翻译：

1. AMC 8, 2016 年, 第 20 题: a 和 b 的最小公倍数是 12, b 和 c 的最小公倍数是 15。 a 和 c 的最小公倍数的最小可能值是多少?

2. AMC 8, 2016 年, 第 24 题: 数字 1、2、3、4 和 5 各用于写入五位数字 PQRST 一次。三位数的 PQR 可以被 4 整除, 三位数的 QRS 可以被 5 整除, 三位数的 RST 可以被 3 整除。什么是 P?

3. AMC 8, 2018 年, 第 21 题: 有多少正三位数整数除以 6 后余数为 2, 除以 9 余数为 5, 除以 11 余数为 7?

4. AMC 8, 2017 年, 第 24 题: 桑德斯夫人有三个孙辈, 他们经常给她打电话。一个三天打一次电话, 一个四天打一次, 还有一个五天打一次。三人均于 2016 年 12 月 31 日给她打电话。接下来的一年里, 她有多少天没有接到孙辈们的电话?